OMCL: Open-vocabulary Monte Carlo Localization

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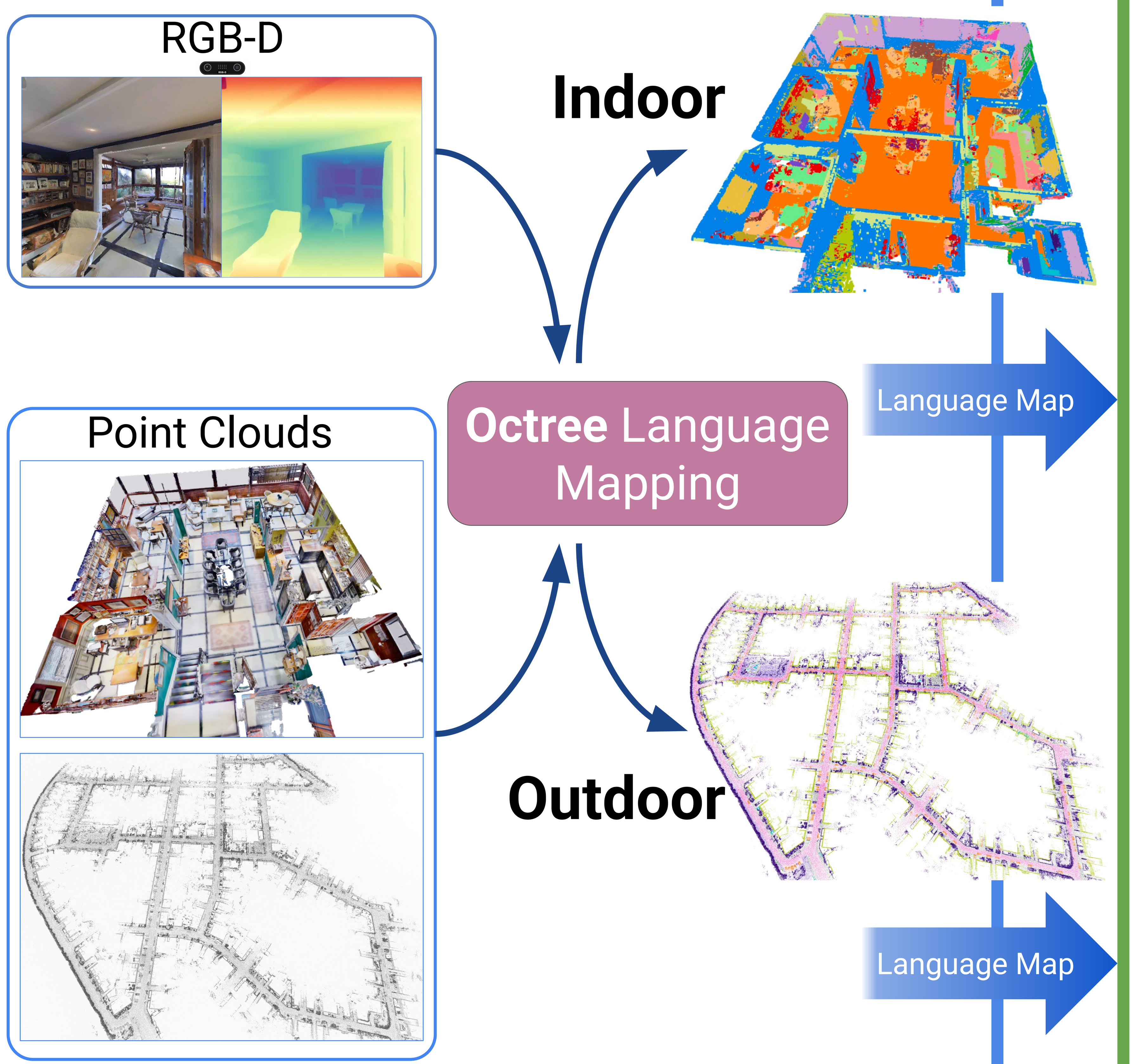
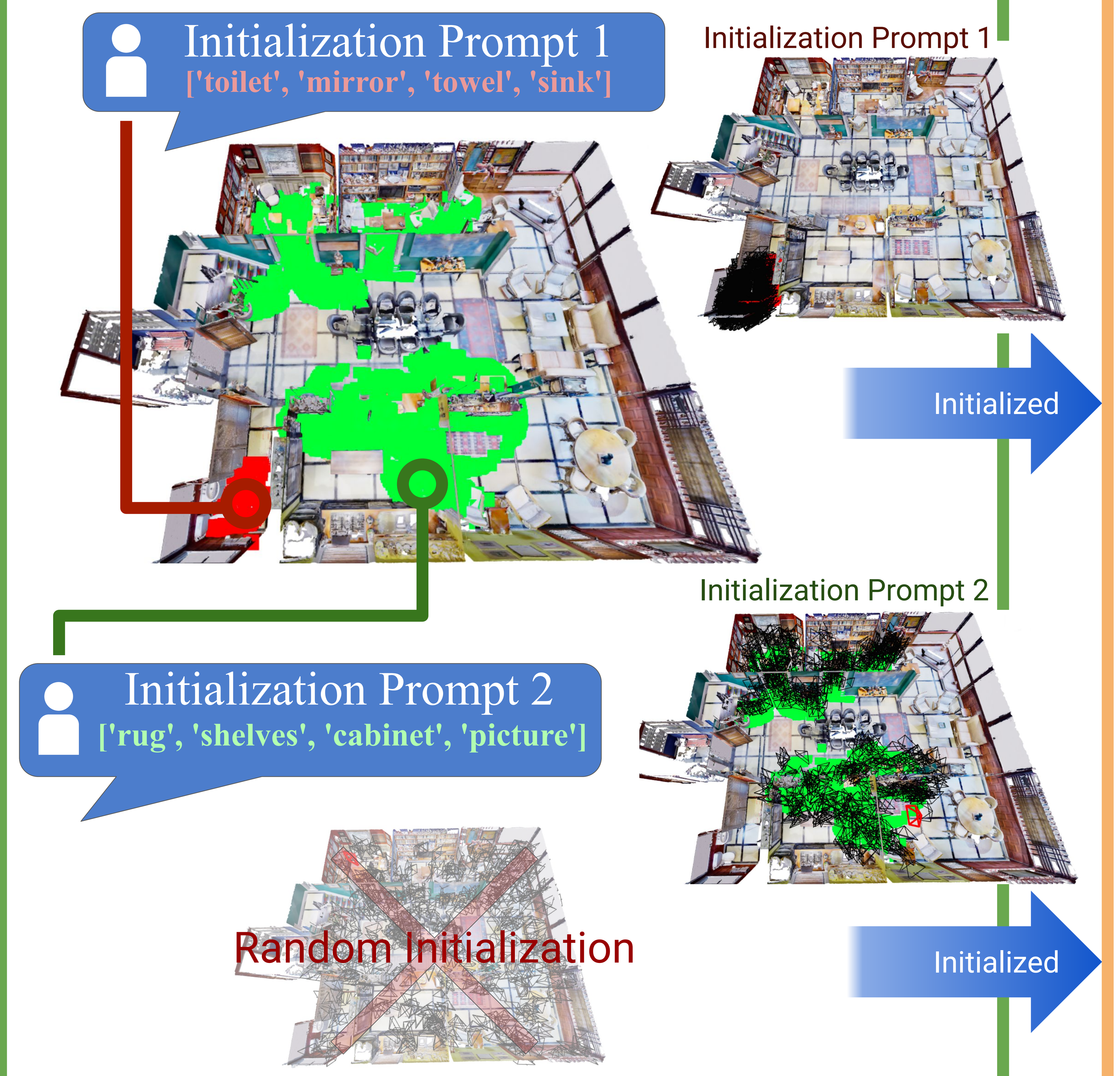
1 Map Construction



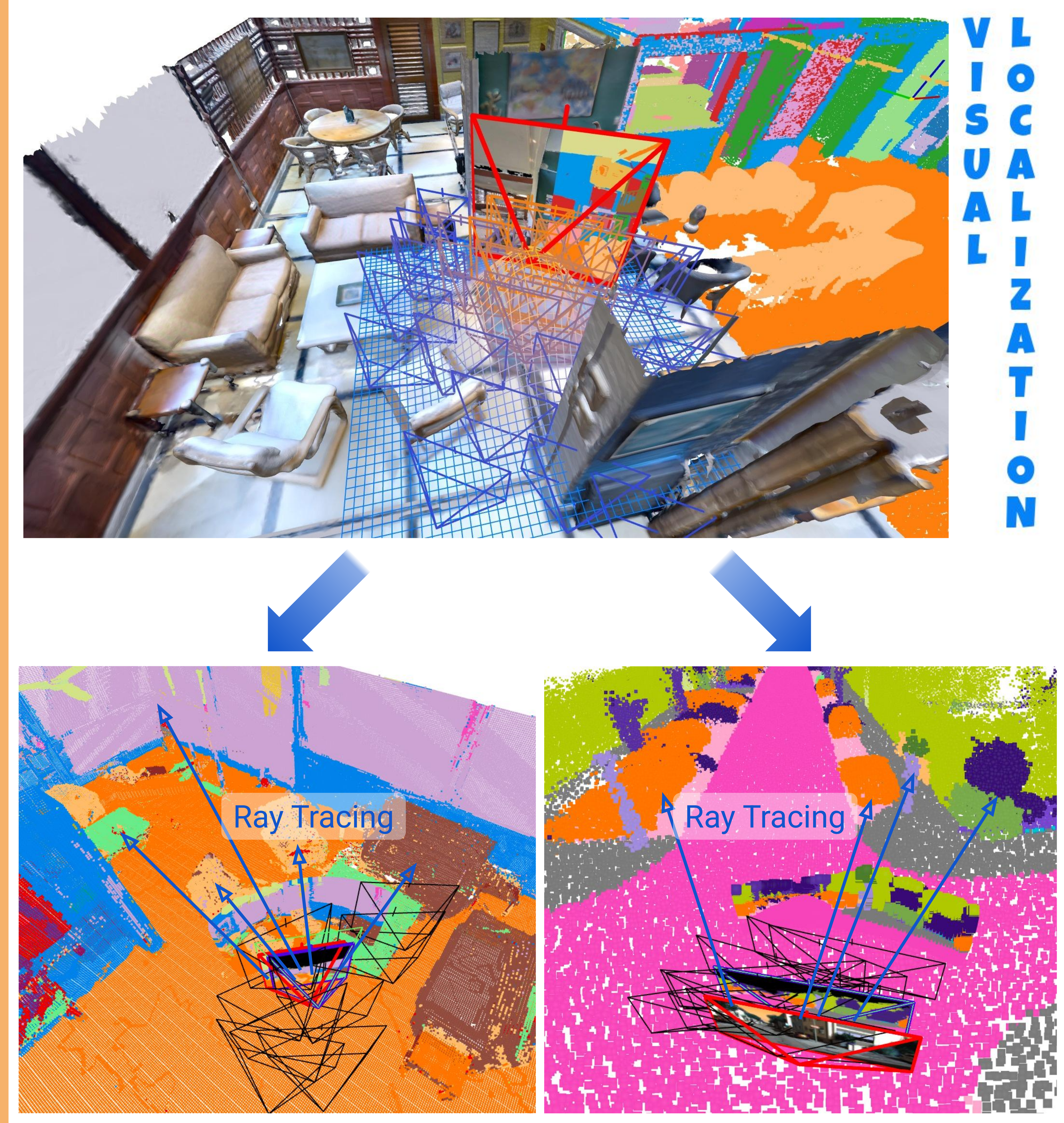
We construct maps from **RGB-D** data or directly convert **existing Point Clouds**, enabling their reuse.

2 Global Initialization



The user prompt allows **non-expert** users to **initialize localization** and **accelerates** its **convergence**.

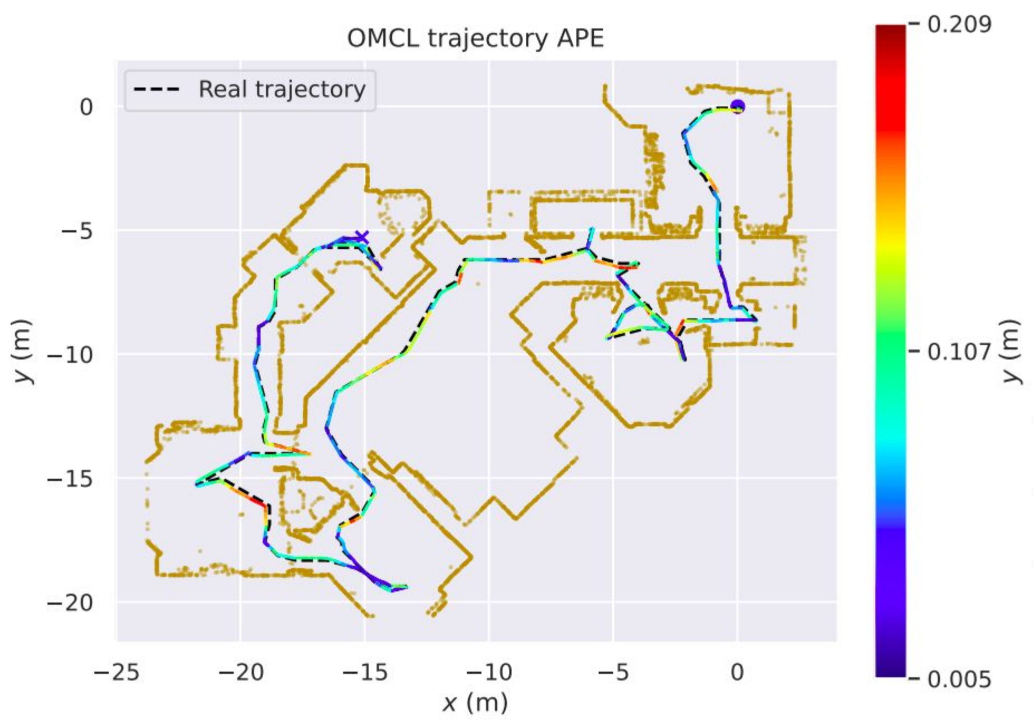
3 Monocular Localization

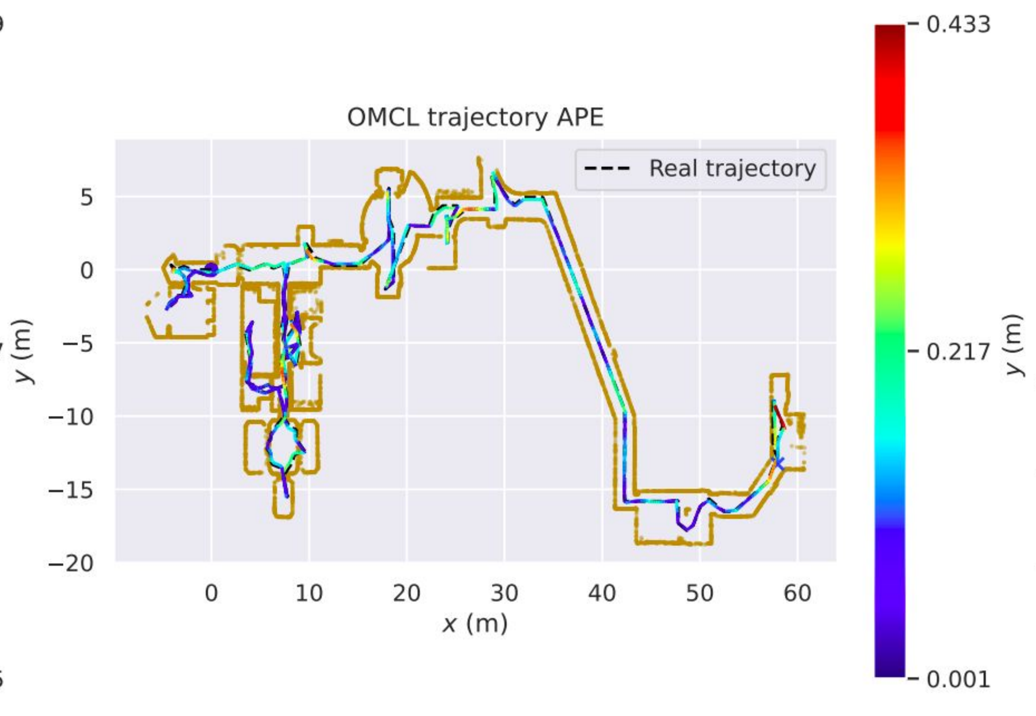


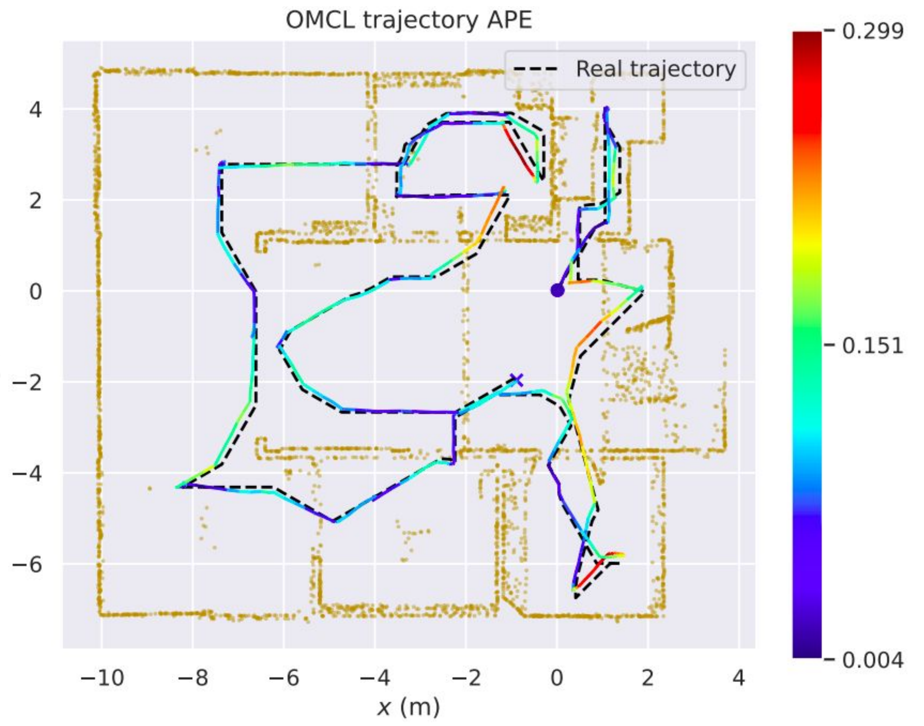
Indoor, RGB+odometry

Outdoor, RGB+odometry

The localization uses only **RGB** and odometry measurements. The particles are weighted based on **observed** and **ray-traced semantics**.







Particles Weights

$$w_i^t = \frac{w_i^{t-1}\max(\mathcal{L}_i, 0)}{\sum_i (w_i^{t-1}\max(\mathcal{L}_i, 0))}$$

$$\mathcal{L}_i = \frac{1}{N}\sum_{j=1}^{N}\frac{\varphi_j \cdot \gamma_j}{\|\varphi_j\|\|\gamma_j\|}$$

φ_j and γ_j are the measured and observed features

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github.com/AIS-Bonn/omcl